

# Satisfacción

**Definición 1 (Satisfacción).** Sean  $A$  una L-estructura,  $a \in A^x$  una evaluación y  $\varphi \in \text{For L}$  una fórmula,  $a$  satisface  $\varphi$  en  $A$  ( $A \models \varphi(a)$ ) recursivamente en la complejidad de  $\varphi$  del siguiente modo:

- (i) Si  $\varphi = \top$ ,  $A \models \varphi(a)$ .
- (ii) Si  $\varphi = t_1 t_2$ ,  $A \models \varphi(a) \iff t_1^A(a) = t_2^A(a)$ .
- (iii) Si  $\varphi = R t_1 \dots t_m$ ,  $A \models \varphi(a) \iff (t_1^A(a), \dots, t_m^A(a)) \in R^A$ .
- (iv) Si  $\varphi = \wedge \varphi_1 \varphi_2$ ,  $A \models \varphi(a) \iff A \models \varphi_1(a)$  y  $A \models \varphi_2(a)$ .
- (v) Si  $\varphi = \neg \psi$ ,  $A \models \varphi(a) \iff A \not\models \psi(a)$ .
- (vi) Si  $\varphi = \exists y \psi$ ,  $A \models \varphi(a) \iff$  existe  $b \in A$  tal que  $A \models \psi(a; b/y)$ .

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